



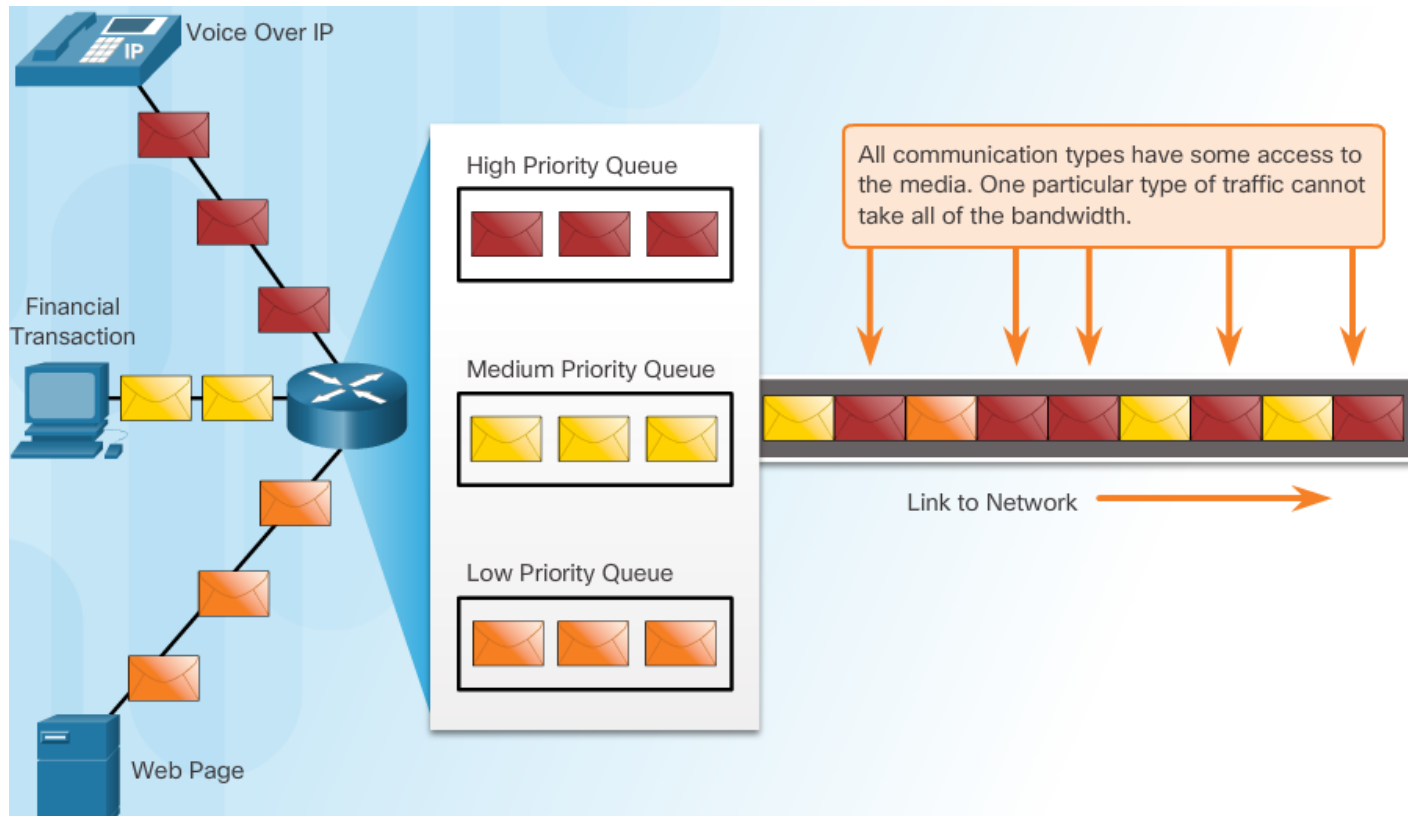
■ Hvad tænker I om begrebet QoS?



Network Transmission Quality

■ Prioritering af trafik

- Når pakker skal ligge i kø skabes delay, fordi nye pakker ikke kan videresendes inden de tidligere pakker er behandlet.
- Hvis antallet af pakker der venter i kø bliver ved med at forøges, kan enheden (switch/router) ikke blive ved med at have plads til at opbevare pakkerne og pakkerne droppes!





Network Transmission Quality

- Hvorfor opstår der kø? (hvis der gør!)
- Bandwidth, Congestion, Delay, and Jitter
 - **Delay** is the time it takes for a packet to travel from the source to the destination.
 - **Jitter** is the variation in the delay of received packets.
- Packet Loss
 - When **congestion** occurs, network devices such as routers and switches can **drop packets**.
 - Packet loss is a very **common cause of voice quality problems** on an IP network.
 - In a properly designed network, packet loss should be near zero.
 - Network engineers use QoS mechanisms to **classify packets** for zero packet loss.



Traffic Characteristics

Forskellige krav til forskellige typer trafik:

■ Voice

- Voice is very **sensitive to delays** and dropped packets; there is no reason to re-transmit voice if packets are lost. Voice can tolerate only a certain amount of latency, jitter, and loss without noticeable effects.
- **Voice packets must receive a higher priority than other types of traffic.**

■ Video

- Video can **tolerate a certain** (higher than Voice) **amount of latency**, jitter, and loss without any noticeable affects.

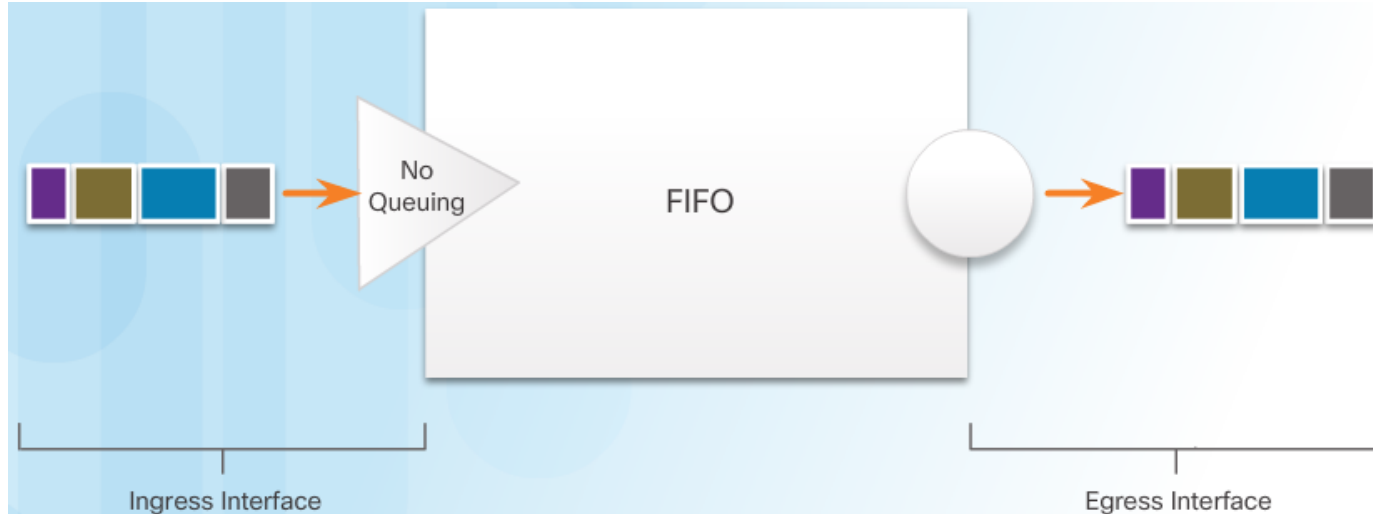
■ Data

- Data applications that have no tolerance for data loss, such as email and web pages, use TCP to ensure that, if packets are lost in transit, they will be resent.
- **Data traffic is relatively insensitive** to drops and delays compared to voice and video.



Queueing Algorithms

- First In First Out (FIFO) = ingen prioritet.
 - FIFO has no concept of priority or classes of traffic and consequently, makes **no decision about packet priority**.
 - FIFO, which is the **fastest method** of queueing, is effective for large links that have little delay and minimal congestion.



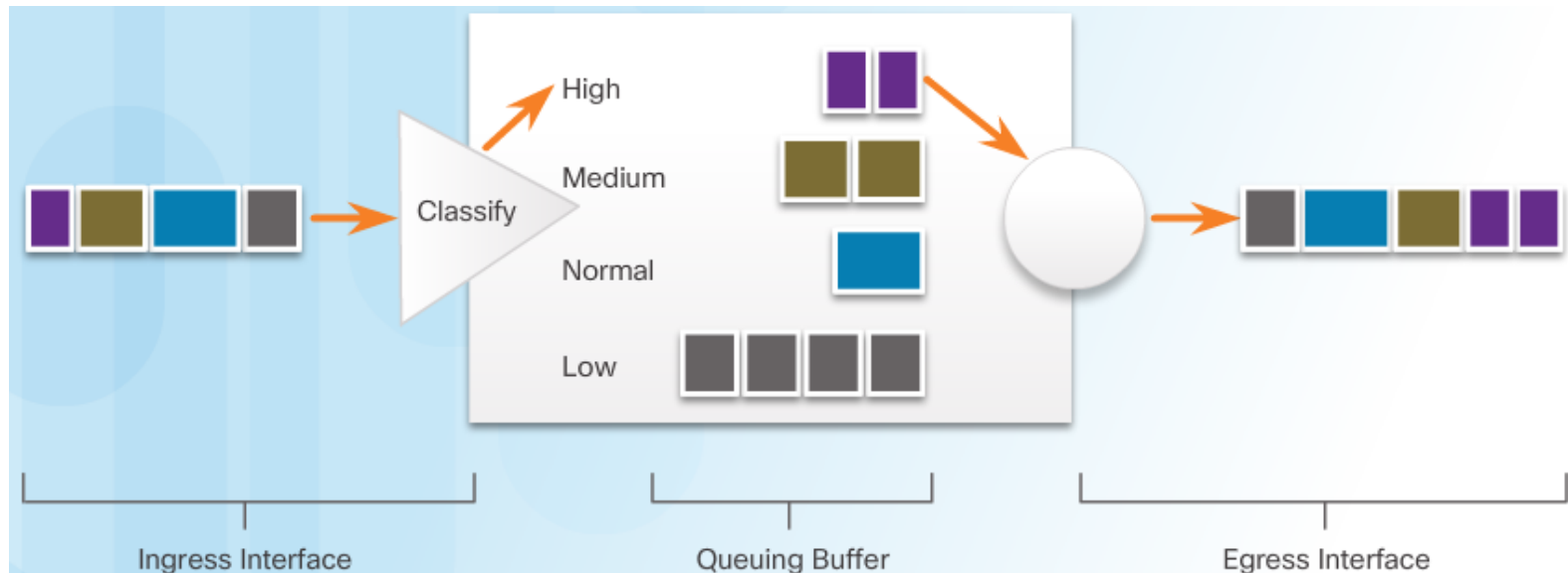


Queueing Algorithms

Weighted Fair Queuing (WFQ) = trafik inddeles i **foruddefinerede klasser**

(hvis vi selv vil definere klasserne, kaldes det Class-Based Weighted Fair Queuing, og det ser vi på næste side)

- An **automated** scheduling method that provides fair bandwidth allocation to all network traffic.
- Applies **priority (= weights)**, to identified traffic and classifies it into conversations or flows. [Klasserne er foruddefinerede, som på tegningen!]
- **!!!! WFQ is not supported with tunneling and encryption** because these features modify the packet content information required.

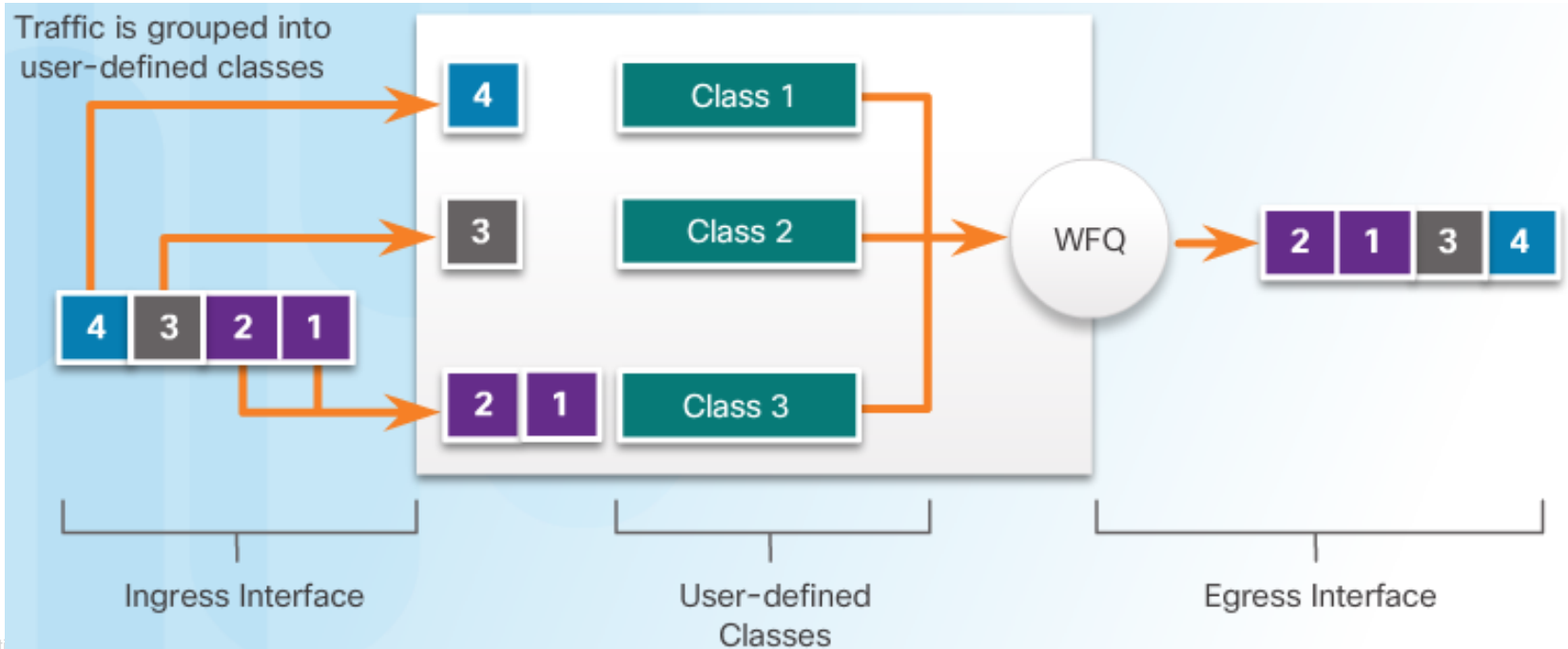




Queueing Algorithms

■ Class-Based Weighted Fair Queuing (CBWFQ)

- Extends the standard WFQ functionality to provide support for **user-defined traffic classes**.
- To characterize a class, you assign it bandwidth, weight, and maximum packet limit.
- You also specify the queue limit for that class, which is the maximum number of packets allowed to accumulate in the queue for the class.

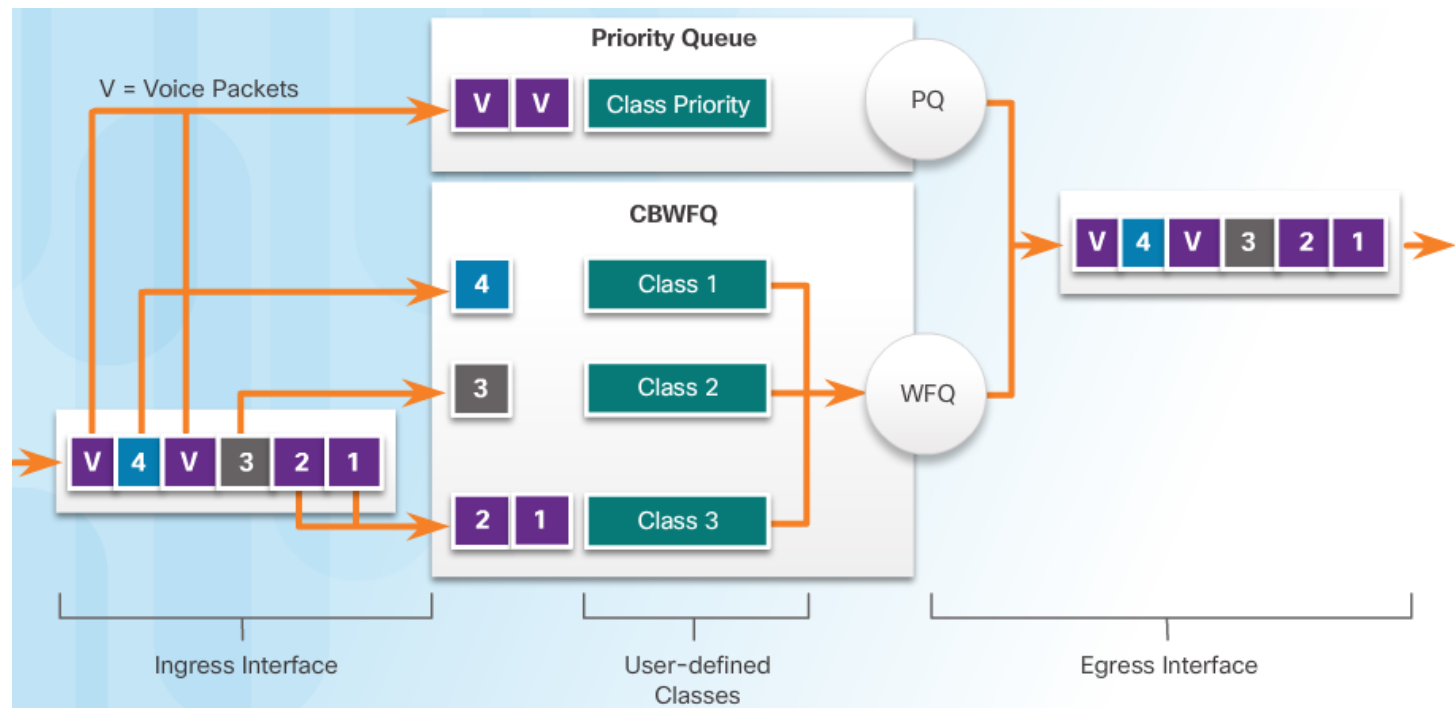




Queueing Algorithms

Low Latency Queuing (LLQ) = noget får absolut førsteret 😊

- LLQ provides strict priority (= har altid førsteret, inden almindelig CBWFQ træder i kraft for resten af trafik typerne).
- LLQ allows delay-sensitive data such as voice to be sent first.





QoS Modeller

Model	Description
Best-effort model	- Not really an implementation as QoS is not explicitly configured. - Use when QoS is not required.
Integrated services (IntServ)	- Provides very high QoS to IP packets with guaranteed delivery. - It defines a signaling process for applications to signal to the network that they require special QoS for a period and that bandwidth should be reserved. - However, IntServ can severely limit the scalability of a network.
Differentiated services (DiffServ)	- Provides high scalability and flexibility in implementing QoS. - Network devices recognize traffic classes and provide different levels of QoS to different traffic classes.

Best-effort = ingen QoS implementeret! (FIFO)

IntServ = Kaldes “integrated” services, fordi enhederne som behandler pakkerne fra afsender til modtager, sender besked i forvejen at der om lidt kommer noget trafik som skal have en speciel behandling. Ikke alle enheder kan genkende denne signaleringsmetode. Duer ikke i store komplekse netværk.

DiffServ = Datapakkerne er opdelt i klasser, som alle enheder ved hvordan skal behandles. Virker også på tværs af store netværk (f.eks. Internettet).
 Altså de klassificerings metoder vi lige har talt om 😊



QoS Implementation Techniques

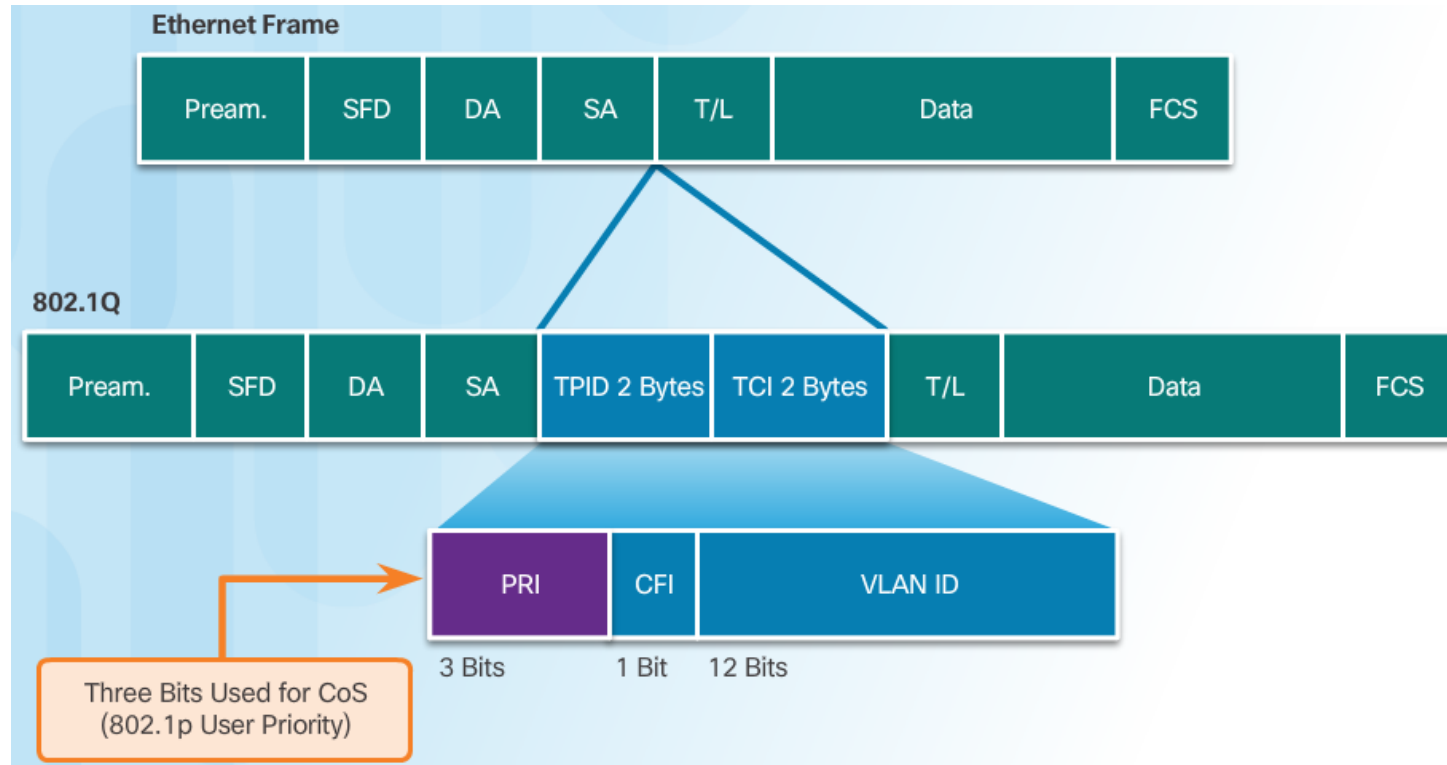
- Classification and Marking
 - Before a packet can have a QoS policy applied to it, the **packet has to be classified.**
 - I praksis betyder det at der udfyldes et felt i L3 og/eller L2 headeren 😊
 - Se i nedenstående skema hvornår man markerer på hhv. **Layer 2 og 3**, hvad markeringsfeltet hedder og hvor meget det fylder.

QoS Tools	Layer	Marking Field	Width in Bits
Ethernet (802.1Q, 802.1p)	2	Class of Service (CoS)	3
802.11 (Wi-Fi)	2	Wi-Fi Traffic Identifier (TID)	3
MPLS	2	Experimental (EXP)	3
IPv4 and IPv6	3	IP Precedence (IPP)	3
IPv4 and IPv6	3	Differentiated Services Code Point (DSCP)	6

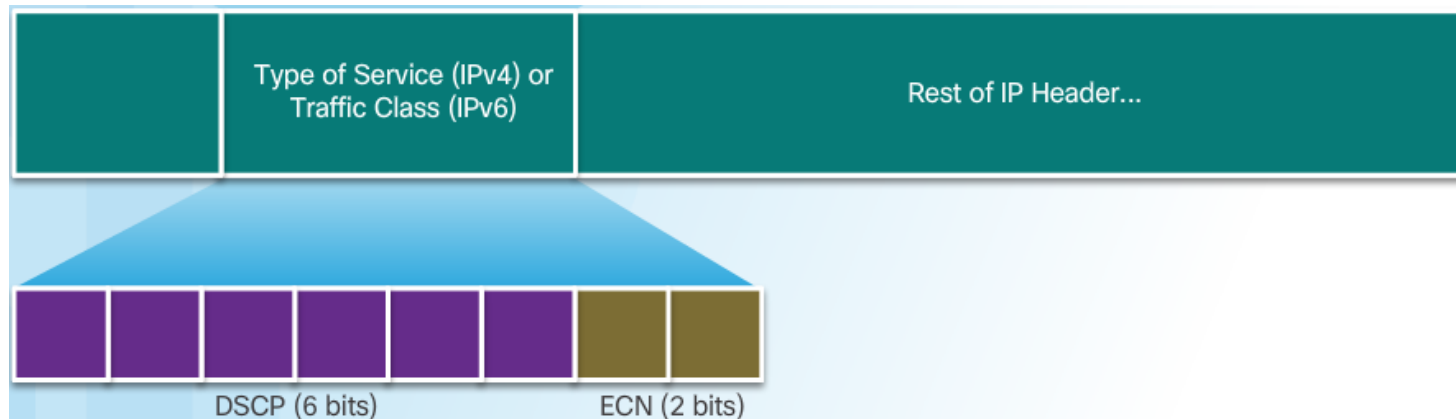


QoS Implementation Techniques

■ Marking at Layer 2



■ Marking at Layer 3

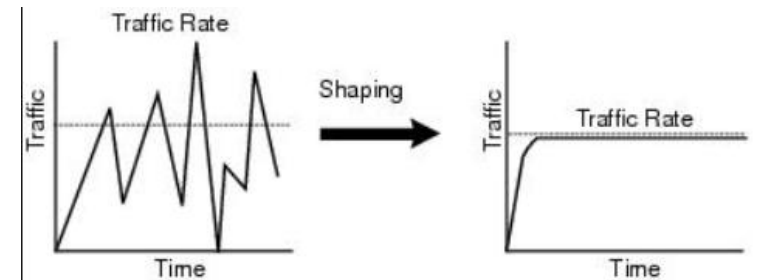




QoS Implementation Techniques

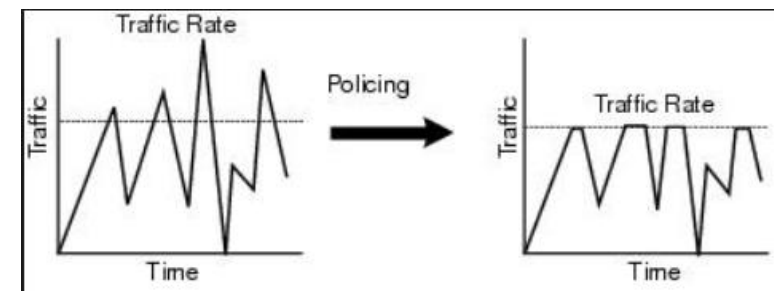
- Trust Boundaries (= der hvor pakkerne bliver klassificeret)
 - Traffic should be classified and marked **as close to its source** as technically and administratively feasible.
- Shaping and Policing

Traffic **shaping** opbevarer udgående pakker i en kø og fordeler trafikken til **senere transmission** (hen over længere tid).



Policing kontrollerer indgående trafik på et interface.

Hvis trafikmængden overstiger den konfigurerede maximum rate, klippes den del af trafikken som overstiger max, og trafikken **droppes** (eller “ommærkes”) i stedet for at videresendes.



>>>Video: MicroNugget_ Using Quality of Service (QoS) Tools<<<< (ca. 5 min.)
[..\..\MicroNugget_ Using Quality of Service \(QoS\) Tools.mp4](..\..\MicroNugget_ Using Quality of Service (QoS) Tools.mp4)

